TrajectoryModel Architecture Description

1 Overview

The TrajectoryModel predicts future vehicle trajectories from two RGB images (current and previous frames). It uses a motion-aware encoder that extracts multi-scale features from each frame, fuses them with temporal differences, and then applies a lightweight transformer decoder to predict the future displacement vectors. An optional auxiliary head can predict a 2D dynamics vector from pooled features. The default input resolution is (B, 3, 360, 640), and the default output is (B, 12, 2) with non-uniform time offsets spanning 3 seconds.

2 Model Components

2.1 Utility Blocks

- **ConvGNAct**: A 2D convolution followed by group normalization and SiLU activation.
- **SEBlock**: Squeeze-and-Excitation block that rescales channels using a global pooling and two-layer MLP.
- **Residual**: Two ConvGNAct layers with a residual connection, optionally followed by SE attention.

2.2 MotionBackbone

A shared-weight CNN extracts features from each frame at two spatial scales.

- **Input**: Image of shape (B, 3, H, W).
- **Architecture**:
  - *Stem*: ConvGNAct (3 to base, kernel=5, stride=2), Residual (base, SE enabled).
  - *Stage1*: ConvGNAct (base to 2 base, stride=2), Residual (2 base, SE enabled).
  - *Stage2*: ConvGNAct (2 base to 3 base, stride=2), Residual (3 base, SE enabled). Output is 1/8 resolution.
  - *Stage3*: ConvGNAct (3 base to 4 base, stride=2), Residual (4 base, SE enabled). Output is 1/16 resolution.
- **Output**: Feature maps at 1/8 and 1/16 resolution.

2.3 MotionFpnEncoder

Fuses motion cues by comparing features between the current and previous frames.

- **Per-scale fusion**: Concatenate current features, previous features, and their difference, then apply a 1x1 convolution to a shared feature dimension.
- **Top-down fusion**: Upsample the 1/16 features and add to the 1/8 features, followed by a ConvGNAct fusion block.
- **Coordinate channels**: Append normalized x/y coordinate maps in the range [-1, 1].
- **Output**: Feature map of shape (B, feat_dim + 2, H/8, W/8). For 360x640 input, this is 45x80.

2.4 VectorTransformerDecoder

The decoder predicts displacement vectors in parallel using a transformer decoder.

- **Memory**: The encoder feature map is projected with a 1x1 convolution and flattened to a sequence of length H W, followed by LayerNorm.
- **Queries**: A learned query embedding of length pred_steps is added to a time-step MLP embedding.
- **Transformer decoder**: A lightweight decoder (default 2 layers, 4 heads) attends to the spatial memory.
- **Output head**: Two-layer MLP that predicts (x, y) per step.
- **Output**: Predicted trajectories of shape (B, pred_steps, 2). Attention maps are not explicitly returned (filled with None).

2.5 TrajectoryModel

The full model integrates the encoder, decoder, and optional auxiliary head.

- **Parameters**: Feature dimension (feat_dim), hidden dimension (hidden_dim), prediction steps (pred_steps=12), optional auxiliary dynamics (default False).
- **Components**:
  - MotionFpnEncoder (outputs feat_dim + 2 channels with coordinate maps).
  - VectorTransformerDecoder (uses feat_dim + 2).
  - Optional auxiliary head: AdaptiveAvgPool2d -> MLP (feat_dim + 2 to 128 to 2).
- **Time offsets**: If not provided, non-uniform offsets are generated for a 3-second horizon (0.1s x 6, 0.25s x 4, 0.7s x 2).
- **Forward**: Returns predicted trajectories, optional auxiliary output, and a placeholder attention list.

3 Summary

The TrajectoryModel combines motion-aware multi-scale encoding with a transformer decoder to predict future vehicle trajectories from image pairs. The design is lightweight, supports non-uniform time steps, and optionally produces a compact auxiliary dynamics output.